Mohit Sharma

Postdoctoral Researcher, Linköping University, Sweden
Ph.D, Indian Institute of Science, Bangalore
Personal Webpage | ORCID | Google Scholar | LiU Profile

Research Profile

My research focuses on scientific visualization and topology-based analysis of multivariate scientific data, with emphasis on multivariate topology, Reeb spaces, Jacobi sets, contour trees, extremum graphs, continuous scatterplots, and topology-driven visual analytics.

I develop scalable, interactive computational methods to extract, simplify, and interpret structures from complex scientific datasets. My work has applications in scientific domains involving spatial and multivariate data, including chemistry, medical imaging, climate science, and scientific simulation data. I am also interested in extending topology-based methods to broader application areas and integrating them with machine learning and AI systems for interpretable analysis of complex scientific data.

Research Output Summary

- 3 journal articles, including 2 in *IEEE Transactions on Visualization and Computer Graphics (TVCG).*
- 4 conference/workshop publications in visualization and topological data analysis venues.
- 3 manuscripts under review and 1 in preparation

Education

Ph.D, Indian Institute of Science, Bangalore 2018–2024
Thesis: *Topological Structures and Operators for Bivariate Data Visualization*, PDF
Defense: March 21, 2024; **CGPA:** 9.3/10; **Advisor:** Prof. Vijay Natarajan

M.Tech, International Institute of Information Technology Hyderabad 2015–2017
CGPA: 9.71/10; Gold Medalist; Dean's List in all semesters

B.Tech, YMCA Institute of Engineering, Faridabad 2007–2011
CGPA: 8.773/10

Academic and Research Experience

Postdoctoral Researcher 2024–present; expected until November 2026
Linköping University, Sweden

- Developing topology-based methods for multivariate and time-varying data analysis, including interactive fiber analysis of bivariate fields using Reeb space.
- Collaborating with researchers in scientific visualization, chemistry, and medical imaging.
- Supervised a Master's research project on summarizing time-varying digital image correlation strain fields.

Doctoral Researcher 2018–2024
Visualization and Graphics Lab, Indian Institute of Science, Bangalore

- Developed topology-based methods for bivariate data visualization, including continuous-scatterplot operators, fiber-surface extraction using Jacobi set, and contour-tree augmentation.
- Applied multivariate topology and continuous scatterplots to the visual analysis of electronic transitions and molecular electronic structure evolution.
- Proposed a simplification method for tracking and analyzing topological features in time-varying scalar fields using the Jacobi set.

Publications

Journal Articles

1. **Continuous Scatterplot and Image Moments for Time-Varying Bivariate Field Analysis of Electronic Structure Evolution**
   Mohit Sharma, Talha Bin Masood, Nanna Holmgaard List, Ingrid Hotz, and Vijay Natarajan
   IEEE TVCG, 2025, DOI.

2. **Continuous Scatterplot Operators for Bivariate Analysis and Study of Electronic Transitions**
   Mohit Sharma, Talha Bin Masood, Signe S. Thygesen, Mathieu Linares, Ingrid Hotz, and Vijay Natarajan
   IEEE TVCG, 2024, DOI.

3. **Jacobi Set Simplification for Tracking Topological Features in Time-Varying Scalar Fields**
   Dhruv Meduri, **Mohit Sharma**, and Vijay Natarajan
   The Visual Computer, 2024, DOI.

Conference and Workshop Publications

4. **Topology-Aware Volume Fusion for Spectral Computed Tomography via Histograms and Extremum Graph**
   Mohit Sharma, Emma Nilsson, Martin Falk, Talha Bin Masood, Lee Jollans, Anders Persson, Tino Ebbers, and Ingrid Hotz
   *TopoInVis, 2025, DOI, **Honorable Mention**.*

5. **Jacobi Set Driven Search for Flexible Fiber Surface Extraction**
   Mohit Sharma and Vijay Natarajan
   TopoInVis, 2022, DOI.

6. **Segmentation Driven Peeling for Visual Analysis of Electronic Transitions**
   Mohit Sharma, Talha Bin Masood, Signe S. Thygesen, Mathieu Linares, Ingrid Hotz, and Vijay Natarajan
   IEEE VIS, 2021, DOI.

7. **On-Demand Augmentation of Contour Trees**
   Mohit Sharma and Vijay Natarajan
   ICVGIP, 2018, DOI.

Posters and Extended Abstracts

8. **Topology-Guided Volume Fusion for Multi-Energy Spectral Computed Tomography**
   Mohit Sharma, Emma Nilsson, Martin Falk, Talha Bin Masood, Lee Jollans, Anders Persson, Tino Ebbers, and Ingrid Hotz
   *VCBM Poster, 2025, DOI, **Best Poster**.*

Manuscripts Under Review and In Preparation

9. **Bringing the Reeb Space to Practice: Efficient Algorithms for Interactive Fiber Analysis of Bivariate Fields**. *Manuscript under review.*

10. **Feature Connectivity Trees: Structural Perspectives for Task-Oriented Analysis**. *Manuscript under review.*

11. **Topological Methods for Multifield Visualization: A Survey**. *Manuscript under review.*

12. **Reeb Space-Driven Feature Tracking in Time-Varying Bivariate Fields**. *Manuscript in preparation.*

Preprints and Technical Reports

13. **Summarizing Time-Varying Digital Image Correlation Strain Fields Using Sankey Diagrams**. Master's student thesis project (supervised). arXiv

Selected Papers for Reprints

- **Continuous Scatterplot and Image Moments for Time-Varying Bivariate Field Analysis of Electronic Structure Evolution**, *IEEE TVCG*, 2025.
- **Topology-Aware Volume Fusion for Spectral Computed Tomography via Histograms and Extremum Graph**, *TopoInVis*, 2025. **Honorable Mention**.
- **Jacobi Set Simplification for Tracking Topological Features in Time-Varying Scalar Fields**, *The Visual Computer*, 2025.
- **Continuous Scatterplot Operators for Bivariate Analysis and Study of Electronic Transitions**, *IEEE TVCG*, 2024.

Teaching and Mentoring Experience

Indian Institute of Science, Bangalore

- Teaching Assistant for courses: Algorithms and Programming; Computational Geometry and Topology; Graphics and Visualization — tutorials, problem-solving sessions, assignments, and evaluation.

International Institute of Information Technology Hyderabad

- Teaching Assistant, Advanced Problem Solving — tutorials, coursework support, and evaluation.

Linköping University

- Master's thesis project (supervised): *Summarizing Time-Varying Digital Image Correlation Strain Fields Using Sankey Diagrams.* arXiv

Teaching Interests

Data Structures and Algorithms; Computer Graphics and Scientific Visualization; Topological Data Analysis and Computational Topology.

Industry Experience

Directi, Platform Engineer Jun 2017–Jan 2018
Developed a machine-learning model for revenue prediction in online advertising.

Adobe, Intern Jun 2016–Jul 2016
Developed a customizable Django-based dashboard for scheduled computation and visualization.

Amazon, Software Development Engineer Jan 2014–Jun 2015
Worked on scalable catalog-quality systems supporting customer-feedback-driven product-data correction.

MediaTek, Software Engineer II Aug 2011–Jan 2014
Developed and optimized mobile applications and widgets for low-resource mobile phones.

Awards and Recognitions

- Best Poster Award, Eurographics Workshop on Visual Computing for Biology and Medicine (VCBM), 2025.
- Honorable Mention, IEEE Workshop on Topological Data Analysis and Visualization (TopoInVis), 2025.
- Gold Medalist, M.Tech. in Computer Science and Engineering, IIIT Hyderabad.
- Dean's List in all semesters during M.Tech., IIIT Hyderabad.

Collaborations

Interdisciplinary collaborations with researchers in scientific visualization, chemistry, medical imaging, and climate science at the Indian Institute of Science, Linköping University, KTH Royal Institute of Technology, and the University of Leeds.

Professional Service

- Reviewer for IEEE VIS and ACM/IEEE Supercomputing Conference (SC).
- Vice Chair, ACM Student Chapter, IISc Bangalore.

Date: 22 May 2026
Place: Norrköping, Sweden